Fitzone : Vision-base Gym Monitoring and Assistance

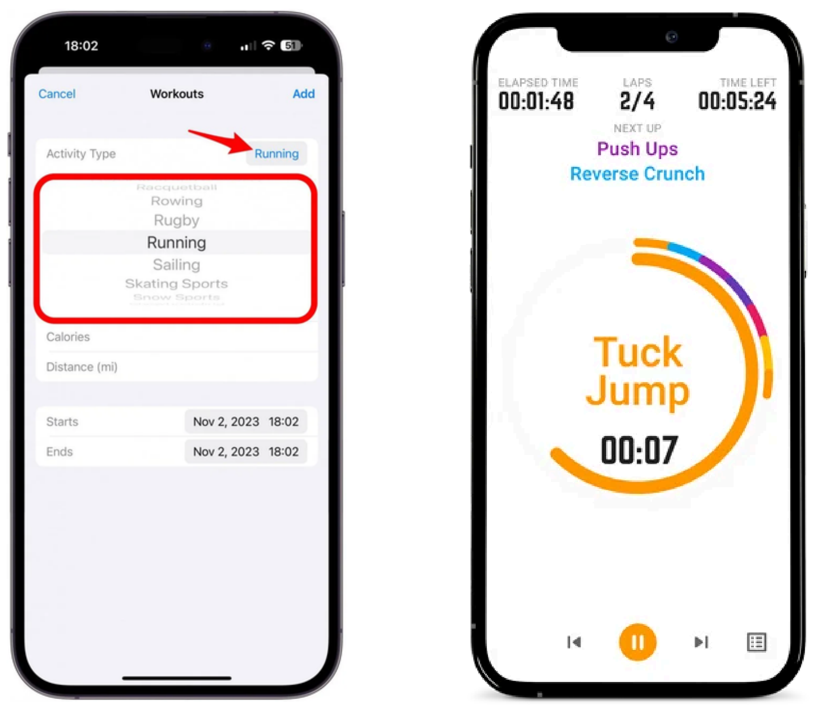
SESSION 2022-26

Abstract

FitZone is an AI-powered smart gym system that automates attendance using face recognition and tracks exercise performance through pose estimation. It integrates a real-time chatbot within the mobile application to guide users during workouts and provides YouTube video for learning correct exercise techniques. A mobile app delivers personalized insights and progress tracking, while a scalable backend and admin dashboard ensure efficient data management and monitoring.

Problem Statement

- Manual workout logging is time-consuming and disrupts user experience.
- Timer-based apps record inaccurate data without verifying exercise performance.
- Lack of automated tracking with personalized guidance and engagement features.



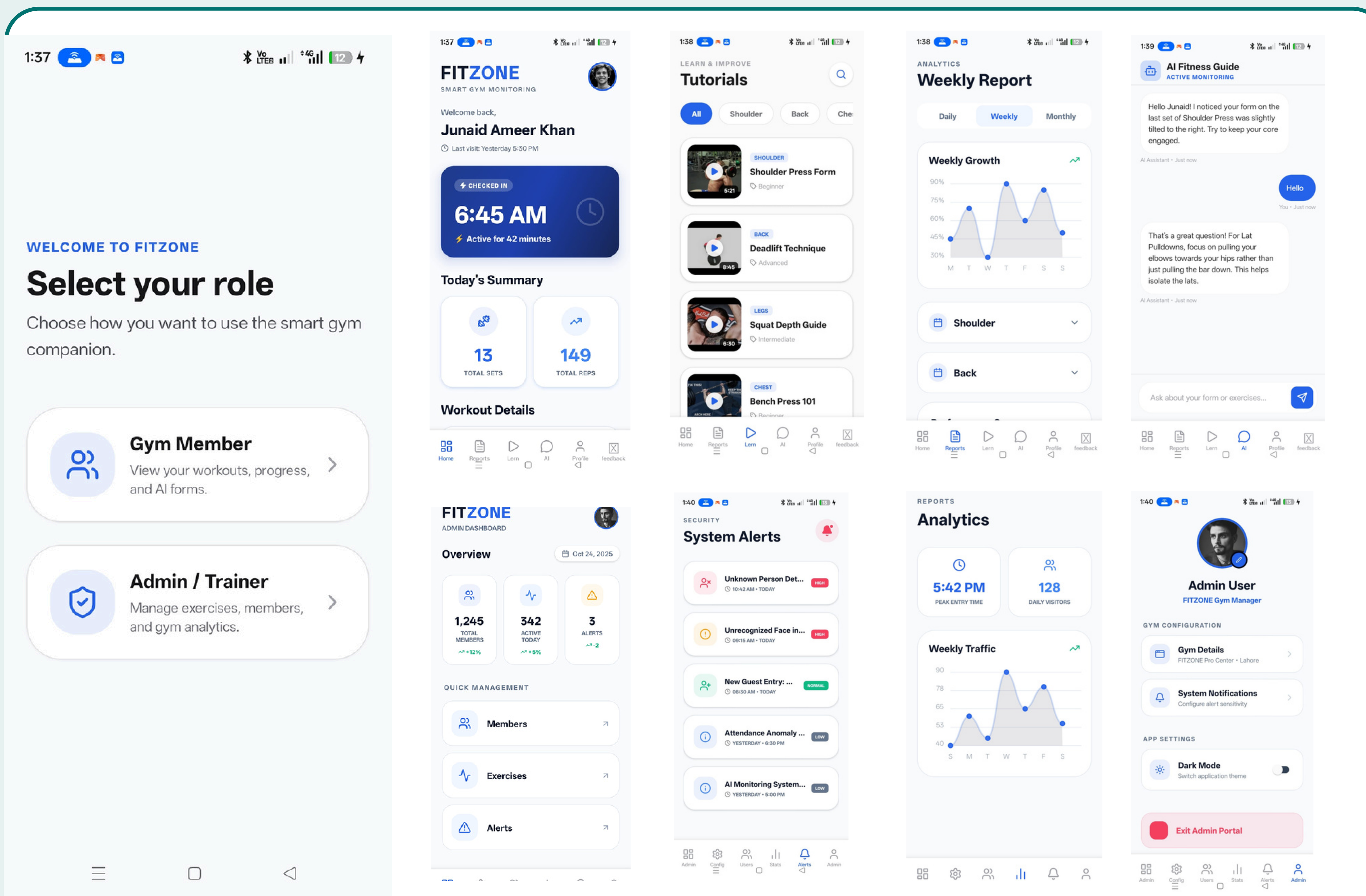
Design and Solution

- Automated exercise tracking using computer vision and AI
- Face recognition system for secure and seamless attendance
- Accurate repetition counting through real-time pose estimation
- Real-time chatbot providing personalized workout guidance
- YouTube video integration for proper exercise learning
- Detailed reports with daily, weekly, and monthly insights
- Gamification features like levels to enhance user motivation
- Scalable backend with mobile app and admin dashboard integration

Took & Technologies

React
GitHub
FastAPI
Firebase

User and Admin Screen



Conclusion

FitZone presents a smart and efficient solution to modern fitness challenges by combining computer vision, artificial intelligence, and user-centered design. It eliminates manual effort, ensures accurate workout tracking, and enhances user engagement through personalized guidance, chatbot assistance, and video learning. With advanced analytics and gamification, FitZone not only improves workout quality but also motivates users, making it a scalable and innovative system for next-generation smart gyms.

Department of Computer Science

Group Members: Junaid Ameer Khan, Zaheer Abbas

Supervisor: Dr. Khawar
Co-supervisor: Mr. Abdul Rafay
Industrial Partner: Tera Data